BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, ABBOTTABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2023

MATHEMATICS

Past Paper 2023 • Paper Code: PR - XII (01) (23)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 20 Min | **Maximum Marks:** 20 | **Paper Code:** PR - XII (01) (23)

ہر سوال کے سامنے چار دائرے دیئے گئے ہیں، صحیح جواب والا دائرہ بھریں۔ دائروں کو کاٹنے، (بھرنے) کے لیے نیلے رنگ کا پین استعمال کریں۔ جواب میں ایک سے زائد دائرے بھرے ہوں تو جواب غلط تصور ہوگا۔

Q1. Domain of function f(x)=|x| is___.

(A) (0,∞) (B) (-∞,∞)

(C) (-∞,0) (D) (0,1)

Q2. The function f(x)=√(x-3) is continuous for all x and___.

(A) x>3 (B) x≤3

(C) x<3 (D) x≥3

Q3. lim(x→0) $(e^x-1)/x$ = ___.

(A) 0 (B) e

(C) 1 (D) x

Q4. d/dx(cos x) = ___.

(A) sin x (B) -sin x

(C) tan x (D) sec x

Q5. If $f(x)=2x^3-3x^2+4$ then f'(2)=___.

(A) 10 (B) 12

(C) 14 (D) 8

Q6. $\lim(x\to0)\ \tan x/x$ = ___.

(A) 0 (B) 2

(C) 1 (D) 3

Q7. If y=sin x then y''' = ___.

(A) sin x (B) -sin x

(C) cos x (D) -cos x

Q8. The graph of a function f(x) is concave downward on (a,b) where:

(A) f''(x)>0 (B) f''(x)<0

(C) f''(x)=0 (D) None of these

Q9. A function f(x) is decreasing if___.

(A) f'(x)>0 (B) f'(x)<0

(C) f'(x)=0 (D) f''(x)=0

Q10. $\int_0^1 x^2 dx$ = ___.

(A) 0 (B) -1/3

(C) 2/3 (D) 1/3

Q11. $\int \cot x\ dx$ = ___.

(A) ln|cos x|+c (B) ln|sin x|+c

(C) tan x+c (D) ln|tan x|+c

Q12. The slope of the line y-1=0 is___.

(A) -1 (B) 0

(C) 1 (D) None of these

Q13. The distance between two points A(3,-2) and B(-1,-5) is___.

(A) 10 (B) 11

(C) 5 (D) 4

Q14. The center of circle $x^2+y^2+2gx+2fy+c=0$ is___.

(A) (g,f) (B) (-g,-f)

(C) (g,-f) (D) (-g,f)

Q15. The parabola $y^2=4px$, if p<0 then it is open___.

(A) Down (B) Up

(C) Left (D) Right

Q16. If the eccentricity e<1, the conic is called___.

(A) Ellipse
(B) Parabola
(C) Circle
(D) Hyperbola

Q17. $y^2/25 - x^2/4 = 1$ is an equation of___.

(A) Circle
(B) Parabola
(C) Ellipse
(D) Hyperbola

Q18. Order of the differential equation $((d^3y/dx^3)^2 - (d^2y/dx^2)^4 + dy/dx + y - 0)$ is___.

(A) 4
(B) 3
(C) 2
(D) 1

Q19. $f(x,y) = (\sqrt{x} + \sqrt{y})/(x+y)$ is a homogenous function of degree___.

(A) -1/2
(B) 1/2
(C) 0
(D) 1

Q20. $\int_0^1 e^{x^2} dx$ has___.

(A) One actual solution
(B) More than one actual solutions
(C) No actual solution
(D) Both A and B

SECTION-B

Paper Code: PR - XII (01) (23)

Note: Time allowed for Section-B and Section-C is 2 Hours and 40 minutes.

Q2. Answer any TEN parts. Each part carries FIVE marks.

1) Find the inverse function of f(x) = (3x+1)/5.
2) Evaluate $\lim(\theta\to0) (\tan\theta - \sin\theta)/\sin^3\theta$
3) Use first principal rule to determine the derivative of $f(x) = 5x^2 - 3x$?
4) Find dy/dx by using any suitable rule of differentiation y = x²e^sin(x)
5) If f(x) = ln(4x+7), find f^iv(x).
6) Evaluate lim(x→0) [sin(3t)/3t i + ln(2+t) j]
7) Find the Maclaurin series expansion for the function $f(x)=\sin^2 x$.
8) Evaluate $\int x^2 \cos x\, dx$
9) Use the integration by parts toe valuate $\int x^2 \cos x\, dx$.
10) Find the centroid of the triangle ABC whose vertices are A(3,5), B(4,6) and C(3,-1).

11) Find the equation of parabola with vertex (-3,4) and passes through (5,11) and its line of symmetry is horizontal.

12) If $f(x,y)=x^2y+xy^2$, then find fx and fy.

13) Solve the initial value problem $dy/dx = 1/x^2$, $y(2)=0$

SECTION-C

> Note: Attempt any THREE questions. All questions carry equal marks.

Q3.

(a) Find dy/dx if $y = ((2x-5)/(x-4))^4$

(b) Find the angle θ from the line $2x+4y-10=0$ to the line $5x-3y+1=0$

Q4.

(a) Find the equation of tangent to the circle $x^2+y^2=9$ which makes an angle of 60° with x-axis?

(b) Find the critical value of $f(x) = x^2/(x-2)$

Q5.

(a) Find the $\int(\pi/2 \text{ to } \pi) \cos(x/2 + \pi)\, dx$

(b) Find an equation of a circle which passes through the points (1,2), (3,-4) and (5,-6).

Q6.

(a) Solve the differential equation $x\, dy/dx = x + y$

(b) Find the equation of hyperbola with vertices are at (5,0) and (-5,0) and one focus at (-7,0).